

4. (12 pts.) Abstract Data Types:

Choose the best abstract data type (List, Set, Map, PQ, Queue, Stack) given the description of the desired data structures below. Be as specific as possible (don't answer List if a Queue is applicable). Also show what types would be used as template arguments (e.g. `map<string, int>` or `stack<double>`). If multiple "best" options exist, choose either. Give a **SHORT** 1-2 sentence justification for your choice (don't waste time on a long explanation...it won't help you get any more credit than a short answer).

4.1. An ADT to support the following: Given a city, retrieve the zip codes that are represented in the city's boundaries. (*Again, write your answers directly in Gradescope*)

4.2. Normally customer service centers process customer calls in order of arrival. Research at USC allows voice analysis to understand the frustration level and emotion of a caller. Thus, a new system orders customers based on their frustration level. This requires changing the customer service software from using a _____ ADT (in the normal system) to a _____ ADT in the new system. No justification necessary (just fill in the blanks).

4.3. An ADT to store each of the individual items in a grocery purchase as they are scanned by the electronic reader.

4.4. An ADT to store the latest changes to **ALL** open documents in a word processor so that the user can, given the document name, undo the latest changes for each document. You may assume a "change" is just a string.